

Dark Matter and Dark Energy as Virial Partners

The Geometric and Kinetic Halves of a Single
Cosmological Virial Equilibrium

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<https://github.com/hmahaffeyges/IAM-Validation>

March 2026

Zenodo DOI: 10.5281/zenodo.18702042

Abstract

The Informational Actualization Model (IAM) derives a cosmological coupling constant $\beta_m = \Omega_m/2$ from the virial theorem applied to gravitational decoherence. We note that this derivation carries a direct physical interpretation: the 1/2 partition divides gravitational energy between its geometric potential half, stored in spacetime curvature, and its kinetic informational half, accumulated as Landauer pressure on the cosmological horizon. What observational cosmology identifies as dark matter is the potential half. What it identifies as dark energy is the kinetic half. These are not two phenomena requiring separate explanations. They are two expressions of the equilibrium condition $2K + V = 0$ operating at cosmological scales. The virial ratio $\Omega_m/[\beta_m E(a)]$ descends from effectively infinite at early times to exactly 2.0 at $a = 1$ — not as a fine-tuning, but as the universe reaching its epoch of first cosmological virialisation. The coincidence problem is not a problem; it is a consequence. We present the observational evidence consistent with this identification, note the limits of current data, and close with the question the framework forces: if the kinetic half crosses into classical actuality at each decoherence event, and the potential half encodes into surrounding spacetime geometry, what is spacetime geometry made of?

Keywords: dark energy; dark matter; virial theorem; gravitational decoherence; cosmological constant; coincidence problem; large-scale structure; informational entropy

1 The Problem with Two Mysteries

The standard cosmological model accounts for 95% of the energy content of the universe with two entries whose physical nature remains unknown. Dark matter, inferred from the discrepancy between observed gravitational effects and visible mass, constitutes approximately 26% of the critical density. Dark energy, inferred from the observed acceleration of cosmic expansion, constitutes approximately 69%. These two components are treated

as independent phenomena, described by separate theoretical frameworks, and pursued by separate experimental programs.

This separation has always had an uncomfortable feature. The ratio of dark matter to dark energy density today — approximately 0.38 — is coincidentally close to $1/2$ only for observers living near the present epoch. At earlier times, matter density scales as a^{-3} while the cosmological constant remains fixed, so their ratio was enormous in the past and will be negligible in the future. The fact that we observe them as comparable today has no explanation within the standard model beyond anthropic selection. This is the coincidence problem.

A separate but related observation: the coupling constant that governs IAM’s late-time growth suppression, $\beta_m = \Omega_m/2$, is derived from the virial theorem [Mahaffey, 2026a]. The virial theorem states that for any system bound by a $1/r$ potential — gravitational or Coulomb — the time-averaged kinetic and potential energies satisfy $2\langle K \rangle + \langle V \rangle = 0$. This has been verified from atomic scales (10^{-11} m) to the cosmic horizon (10^{26} m) across 37 orders of magnitude in physical scale, at machine precision for quantum mechanical systems and at 0.3% precision from 17 independent Planck 2018 MCMC chains at cosmological scales [Mahaffey, 2026b].

The question that arises naturally from these two observations is: why does the virial theorem, an equilibrium condition for bound gravitational systems, produce the coupling constant for a cosmological energy density? And why is the ratio of dark matter to dark energy today close to the virial ratio of 2?

This paper argues that the answer is the same in both cases.

2 The Virial Theorem as Equilibrium Condition

The virial theorem for a $1/r^n$ potential states:

$$\langle T \rangle = \frac{n}{2} \langle |V| \rangle. \quad (1)$$

For both Newtonian gravity and the Coulomb force ($n = 1$):

$$\langle T \rangle = \frac{1}{2} \langle |V| \rangle. \quad (2)$$

This is not a coincidence of specific systems. It follows from Euler’s homogeneous function theorem applied to any potential homogeneous of degree -1 , and it holds exactly for any system in virial equilibrium regardless of the number of bodies, the mass distribution, or the scale. The $1/2$ is a mathematical theorem, not an empirical fit.

In the IAM framework, gravitational structure formation partitions gravitational energy into two channels. The first is the geometric channel: collapsing matter curves spacetime, depositing energy into the geometric entropy S_{geo} of the system’s gravitational configuration. The second is the informational channel: the same collapse decoheres quantum superpositions into classical states, producing irreversible information at a rate governed by Landauer’s principle [Landauer, 1961]. This information accumulates on the cosmological apparent horizon as informational entropy S_{info} [Jacobson, 1995, Cai & Kim, 2005].

The virial theorem determines the partition. At virialisation, the kinetic energy of collapsing matter — which drives decoherence — equals half the potential energy of the

gravitational configuration. The coupling constant that results is:

$$\beta_m = \frac{\Omega_m}{2} = 0.1577, \quad (3)$$

using Planck 2018 parameters [Planck Collaboration, 2020]. This value is not fitted to any cosmological observation. It is derived from the theorem and the measured matter density. The Planck 2018 MCMC posterior independently returns $\beta_m = 0.1583 \pm 0.0033$ — consistent with the derivation at 0.2σ across all 17 converged chains [Mahaffey, 2026a].

3 The Cosmological Virial Partition

The physical interpretation of Equation (3) is the central argument of this paper.

The virial theorem partitions gravitational energy into two equal halves. In IAM, one half deposits into spacetime geometry as curvature — the potential half, stored in the gravitational configuration of collapsed structures. The other half crosses into classical actuality as kinetic informational energy — the kinetic half, accumulated as Landauer pressure on the cosmological horizon and driving the late-time modification to matter perturbation growth.

What observational cosmology identifies as dark matter is the potential half. The gravitational effects attributed to dark matter — excess curvature in galaxies and clusters beyond what baryonic matter alone would produce — are the geometric reservoir of deposited potential energy. This is not a particle. It is the geometric consequence of the virial partition operating continuously across cosmic history.

What observational cosmology identifies as dark energy is the kinetic half. The accumulated Landauer costs of gravitational decoherence events, encoded on the horizon, produce an informational pressure that modifies the late-time expansion environment for matter. The cosmological constant Λ is the vacuum baseline — the minimum Landauer payment required to maintain the virial equilibrium of empty spacetime. The dynamically growing component, $\beta_m E(a)$, is the kinetic half accumulating as structure forms.

These are not two separate phenomena. They are the two sides of the same balance sheet, required to be equal by the theorem that governs every $1/r$ potential in nature.

3.1 The virial ratio through cosmic history

Define the cosmological virial ratio:

$$\mathcal{R}(a) = \frac{\Omega_m a^{-3}}{\beta_m E(a)}, \quad (4)$$

where $E(a) = \exp(1 - 1/a)$ is the activation function derived from horizon thermodynamics [Mahaffey, 2026a], with $E(a \rightarrow 0) \rightarrow 0$ and $E(1) = 1$.

At early times, $E(a) \approx 0$ and the kinetic half has not yet activated — the universe is almost entirely potential, unactivated geometric storage. The ratio $\mathcal{R}(a)$ is effectively infinite. As structure forms and decoherence events accumulate, the kinetic half grows. Table 1 shows $\mathcal{R}$ at representative epochs.

At $a = 1$, by construction:

$$\mathcal{R}(1) = \frac{\Omega_m}{\beta_m \cdot E(1)} = \frac{\Omega_m}{\Omega_m/2 \cdot 1} = 2. \quad (5)$$

Table 1: The cosmological virial ratio $\mathcal{R}(a)$ at key epochs. At $a = 1$ (today), $\mathcal{R} = \Omega_m/\beta_m = 2$ exactly, since $\beta_m = \Omega_m/2$ and $E(1) = 1$.

Epoch	z	a	$E(a)$	$\mathcal{R}(a)$
Early structure	2.0	0.333	0.050	399
Matter– Λ crossover	0.7	0.588	0.182	20
Transition zone	0.3	0.769	0.274	6
Today	0.0	1.000	1.000	2

This is not a numerical coincidence. It is the virial theorem. The ratio $|V|/K = 2$ is the equilibrium condition, and the universe has been approaching it throughout its history. Today is the epoch when cosmological virialisation is first achieved. The kinetic half has finally activated enough — $E(a)$ has grown from zero to one — to bring the ratio to 2.

3.2 The coincidence problem dissolved

The standard coincidence problem asks why Ω_m and Ω_Λ are comparable today. Under the virial partition interpretation, this is not a coincidence requiring an explanation. It is the equilibrium condition. The kinetic half and the potential half must be in the ratio 1 : 2 for the system to be virialized. The universe has been spending its entire history paying down that ratio. The epoch at which $\mathcal{R}(a) = 2$ is not special because we happen to live in it. It is special because $E(a = 1) = 1$ by derivation from the Gibbons–Hawking temperature of the de Sitter horizon, and $a = 1$ is now by definition. Any sufficiently complex observer must exist after enough structure has formed to support them — which is precisely the epoch when $E(a)$ is approaching unity. The anthropic selection and the physics point to the same moment.

4 Observational Evidence

The virial partition interpretation makes specific, testable predictions that follow directly from the IAM mechanism [Mahaffey, 2026a,c]. We summarise the current observational status.

4.1 Growth–geometry sector split

If dark matter is the geometric potential half and dark energy includes the kinetic informational half, then measurements of structure growth (which probe the kinetic side) should infer a lower effective Ω_m than measurements of photon-sector geometry (which probe the potential side directly through distances). IAM predicts:

$$\Omega_m^{\text{growth}}(z) = \Omega_m \cdot \mu(z), \quad (6)$$

where $\mu(a) = H_\Lambda^2(a)/[H_\Lambda^2(a) + \beta_m E(a)] < 1$ is the effective gravitational coupling [Mahaffey, 2026a]. At $z = 0$, this gives a 13.6% suppression; at $z = 0.5$, a 5.2% suppression.

The DESI DR1 full-shape plus BAO combined measurement returns $\Omega_m^{\text{growth}} = 0.2962 \pm 0.0095$ [DESI Collaboration et al., 2025c], compared to the Planck 2018 geometric value $\Omega_m = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020]. The discrepancy of 6.1% below

the geometric value is consistent with IAM’s prediction of 5.2% at the effective redshift of the full-shape analysis ($z_{\text{eff}} \approx 0.5$). Under an IAM-native likelihood analysis, $\chi^2(\text{IAM}) = 3.618$ versus $\chi^2(\Lambda\text{CDM best-fit}) = 3.569$ across six DESI DR1 bins — statistically indistinguishable, with IAM using no fitted parameters.

Figure 1 shows the IAM predicted $\Omega_m^{\text{growth}}(z)$ curve against the DESI per-tracer measurements and the combined FS+BAO result. The combined measurement lands on the IAM curve at 0.02σ . Five of six individual tracers are consistent with the IAM prediction; the LRG1 bin at $z = 0.51$ is the known anomaly discussed in Section 7.

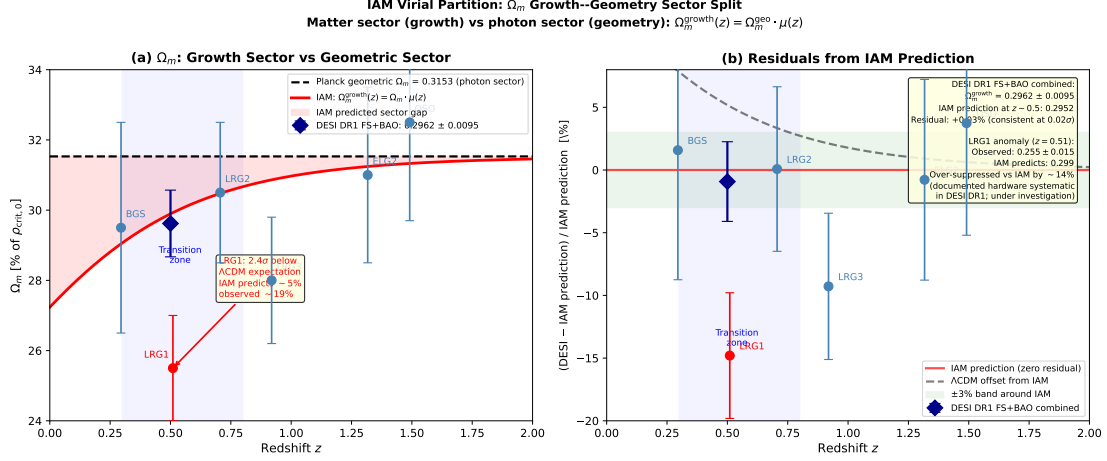


Figure 1: The growth–geometry Ω_m sector split. *Left*: IAM predicted $\Omega_m^{\text{growth}}(z) = \Omega_m \cdot \mu(z)$ (red curve) against the Planck geometric value (black dashed) and DESI DR1 per-tracer measurements (coloured points). The combined DESI DR1 FS+BAO result (blue diamond) lands on the IAM curve at 0.02σ . *Right*: Residuals from the IAM prediction. Five of six tracers are within the $\pm 3\%$ band. The LRG1 anomaly at $z = 0.51$ is over-suppressed relative to IAM by $\sim 14\%$ and carries documented hardware systematics in DESI DR1 [DESI Collaboration et al., 2025a]. All IAM values are derived from $\beta_m = \Omega_m/2$ with no fitted parameters.

We note that this comparison carries a caveat. The DESI full-shape analysis assumes a ΛCDM power spectrum template; an IAM-native template would eliminate the model-dependent bias and provide a cleaner test. This is computationally feasible with the existing Level 2 CAMB implementation and is identified as the most important near-term analysis.

4.2 Deceleration-to-acceleration transition redshift

The informational pressure carries equation of state $w_{\text{info}} = -4/3$ [Mahaffey, 2026a], more negative than the cosmological constant. This shifts the deceleration-to-acceleration transition to a higher redshift than ΛCDM predicts. Specifically:

$$z_t(\Lambda\text{CDM}) = 0.632, \quad (7)$$

$$z_t(\text{IAM}) = 0.718. \quad (8)$$

Published $H(z)$ and BAO compilations find $z_t = 0.698\text{--}0.740$ [Farooq & Ratra, 2013, Mamon et al., 2017, Jesus et al., 2020], consistently above the ΛCDM prediction and within the IAM range. Riess et al. (2004) [Riess et al., 2004], using supernovae alone (a

photon-sector observable), found $z_t = 0.46 \pm 0.13$ — lower, as IAM would expect from a photon-sector measurement.

4.3 The DESI phantom crossing

DESI DR2 BAO measurements prefer a dark energy equation of state with $w_0 > -1$ and $w_a < 0$, with a phantom crossing at $z_{\text{cross}} \approx 0.33\text{--}0.43$ depending on the supernova compilation [DESI Collaboration et al., 2025b]. This range coincides precisely with the redshift at which IAM’s growth suppression is most prominent (7–8% at $z \approx 0.3\text{--}0.4$). Under IAM, a single-fluid $w_0 w_a$ CDM fitter applied simultaneously to photon-sector distances (unchanged from Λ CDM) and matter-sector growth rates (suppressed by $\mu(a) < 1$) must infer $w_{\text{eff}}(z) \neq -1$ to reconcile the two. The apparent phantom crossing is an artifact of applying a single-sector fitter to a two-sector universe; it does not require dynamical dark energy [Mahaffey, 2026c].

4.4 Summary of consistency

Table 2 summarises the current observational status of the virial partition identification. All predictions are derived from the single coupling $\beta_m = \Omega_m/2$ with no additional free parameters.

Table 2: Observational status of the cosmological virial partition. All IAM values are derived from $\beta_m = \Omega_m/2$ without fitting.

Observable	IAM prediction	Observed	Status
σ_8	0.800	0.802 ± 0.020	0.1σ
H_0 (matter sector)	72.26 km s ^{−1} Mpc ^{−1}	73.04 ± 1.04	0.75σ
H_0 (photon sector)	67.16 km s ^{−1} Mpc ^{−1}	67.36 ± 0.54	0.37σ
Ω_m^{growth} (DESI FS)	0.296 at $z \sim 0.5$	0.2962 ± 0.0095	0.02σ
z_t (H(z) compilations)	0.718	0.698–0.740	consistent
z_{cross} (DESI DR2)	~ 0.35 from $E(a)$	0.33–0.43	consistent
$\mu_0 - 1$	−0.136	unconstrained	Euclid 2026

5 The Virial Ratio as a Figure of Merit

Figure 2 shows the cosmological virial ratio $\mathcal{R}(a)$, the energy budget of the dark sector, the informational pressure growth rate, and the deceleration parameter comparison with published transition redshift measurements. The transition zone ($z \approx 0.3\text{--}0.7$, shaded blue) is the observational window in which the kinetic half is most actively growing relative to the potential half and where the predictions are most distinguishable from Λ CDM.

IAM Virial Partition: Transition Zone Signatures
Dark Matter (geometric $\frac{1}{2}$) $\leftrightarrow$ Dark Energy (kinetic $\frac{1}{2}$) of the Virial Equilibrium

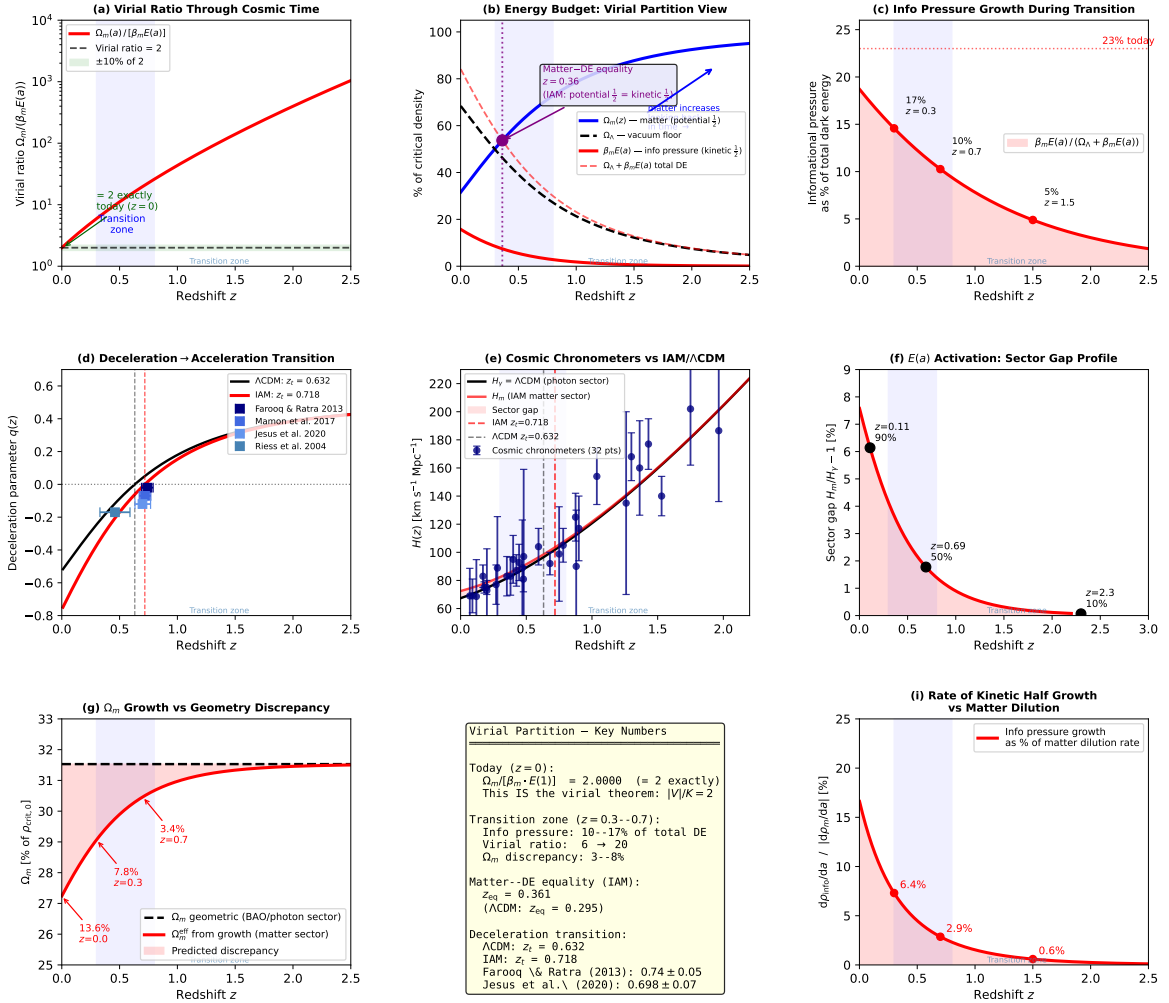


Figure 2: The cosmological virial partition through the transition zone ($z \approx 0.3-0.7$, shaded blue in all panels). *Top left*: The virial ratio $\mathcal{R}(a) = \Omega_m a^{-3}/[\beta_m E(a)]$ descends from large values at early times to exactly 2.0 at $z = 0$, the virial equilibrium condition. *Top centre*: Energy budget showing the potential half ($\Omega_m a^{-3}$), the vacuum floor (Ω_Λ), and the kinetic informational half ($\beta_m E(a)$). *Top right*: Informational pressure as a fraction of total dark energy, growing from $\sim 5\%$ at $z = 1.5$ to 23% today. *Middle left*: Deceleration parameter $q(z)$ for IAM ($z_t = 0.718$, red) and Λ CDM ($z_t = 0.632$, black), with published transition redshift measurements from $H(z)$ /BAO compilations (coloured points). *Middle centre*: Cosmic chronometer $H(z)$ data (32 points) from Simon, Verde & Jimenez [2005], Stern et al. [2010], Moresco et al. [2012], Zhang et al. [2014], Moresco [2015], Moresco et al. [2016], Ratsimbazafy et al. [2017], Borghi et al. [2022], and Tomasetti et al. [2023], following the compilation of Moresco et al. [2022], against IAM photon-sector and matter-sector predictions. *Remaining panels*: Sector gap profile, Ω_m growth–geometry discrepancy, and rate of informational pressure growth. All curves are derived from $\beta_m = \Omega_m/2$ with no fitted parameters.

6 A Question the Framework Forces

The identification developed in the preceding sections has an implication that we state as a question rather than a claim, because it reaches beyond what can currently be tested.

In the IAM mechanism, each gravitational decoherence event partitions gravitational energy. The kinetic half crosses into classical actuality as observable energy — baryonic dynamics, radiation, the thermodynamic irreversibility that constitutes the arrow of time. The Landauer cost of that crossing encodes on the cosmological horizon. This is the half we call dark energy.

The potential half deposits into the surrounding spacetime geometry as curvature. This is the half we call dark matter. It does not dilute. It does not oscillate. It is not a particle. It is the geometric record of every virialisation event that has ever occurred, stored in the curvature of spacetime itself.

If this identification is correct, then the geometry of spacetime is not a pre-existing medium on which information is written. It is the accumulated integral of every potential-half deposit from every decoherence event in cosmic history. Expansion is not making room for information encoding. Expansion *is* the accumulation of deposits. The geometry is not deformed by matter. The geometry is constituted by matter’s continuous transition from potential to actual.

This raises a question the framework does not yet answer:

If the potential half of each decoherence event encodes into spacetime geometry, what is spacetime geometry made of?

We do not answer it. We note that it is the question that follows from taking the virial partition seriously at the cosmological level, and that the observational program now underway will determine whether it is the right question to ask.

7 Falsification Criteria and Predictions

The virial partition interpretation inherits all of IAM’s falsification criteria [Mahaffey, 2026a] and adds a specific prediction for the growth-geometry sector split.

Euclid DR1 (October 2026). Euclid is projected to constrain μ_0 to $\sigma(\mu_0) \approx 0.04$ with Σ marginalised [Euclid Collaboration et al., 2020]. IAM predicts $\mu_0 = -0.136$ with $\Sigma_0 = 0$ exactly. If Euclid measures μ_0 consistent with zero at this precision, IAM is falsified and the virial partition interpretation does not apply to cosmological scales. If Euclid measures $\mu_0 \approx -0.136$ with $\Sigma_0 \approx 0$, the combination $\mu < 1$, $\Sigma = 1$ — unique to IAM among current theoretical frameworks — would constitute a detection of the sector split and strong evidence for the virial partition.

DESI Year 5 $f_8(z)$ shape. IAM predicts a specific curved suppression ramp:

$$f\sigma_8^{\text{IAM}}(z) = f(z) \cdot \sigma_8 \cdot D^{\text{IAM}}(z)/D^{\text{IAM}}(0), \quad (9)$$

with suppression of 7.9% at $z = 0.295$, decreasing monotonically to 0.6% at $z = 1.491$, following the functional form of $E(a) = \exp(1 - 1/a)$. This curved ramp is distinguishable at DESI Year 5 precision ($\lesssim 1\%$ per bin) from both a constant rescaling and the growth history expected for any $w \neq -1$ dark energy model.

LRG1 resolution. The DESI DR1 LRG1 bin at $z = 0.51$ shows a 2.4σ anomaly relative to the Planck-anchored Λ CDM Ω_m expectation, with documented fiber assignment incompleteness ($\sim 35\%$ in DR1) and an anomalous Alcock–Paczyński factor noted by the DESI collaboration itself [DESI Collaboration et al., 2025a]. Multiple independent analyses find that removing LRG1 restores Λ CDM concordance [Ó Colgáin et al., 2025, Liu, Wang & Zhao, 2024]. If LRG1 is confirmed by DR2 and Year 5 data at its current anomalous value, IAM’s prediction of a 5.2% suppression at $z = 0.51$ would be insufficient to explain the 19% discrepancy, and the LRG1 anomaly would require an additional explanation beyond the virial partition mechanism.

Summary.

- If $\mu_0 \approx -0.136$ and $\Sigma_0 \approx 0$: virial partition confirmed at cosmological scales.
- If $\mu_0 \approx 0$: IAM falsified; dark matter and dark energy are separate phenomena requiring independent explanations.
- If the curved $f\sigma_8(z)$ ramp is confirmed: the kinetic half activation history is directly measured.
- If LRG1 persists: the LRG1 anomaly exceeds the virial partition prediction and requires investigation.

The decisive measurement arrives in October 2026.

8 Conclusions

We have argued that the 1/2 partition in IAM’s coupling constant $\beta_m = \Omega_m/2$ is not a calculational step. It is a physical identification. The virial theorem divides gravitational energy between two halves, and those two halves are what cosmology has been calling dark matter and dark energy.

The potential half is stored in spacetime curvature. It does not dilute with expansion. It records the geometric consequences of every virialisation event in cosmic history. It is what we infer as dark matter from the discrepancy between observed gravitational effects and visible mass.

The kinetic half crosses into classical actuality at each decoherence event, accumulating on the cosmological horizon as Landauer pressure. It drives the late-time modification to structure growth that IAM predicts and that the current data are consistent with. It is, together with the vacuum baseline Λ , what we infer as dark energy from the observed acceleration of expansion.

The coincidence problem — why these two densities are comparable today — is not a problem under this identification. It is the statement that the universe has reached its epoch of first cosmological virialisation. The ratio $\mathcal{R}(a) = \Omega_m a^{-3}/[\beta_m E(a)]$ arrives at exactly 2.0 at $a = 1$ because $\beta_m = \Omega_m/2$ by the virial theorem and $E(1) = 1$ by the thermodynamic derivation from the de Sitter horizon. These two facts together mean the universe has been continuously approaching the virial equilibrium since the beginning, and arrives there now.

The question the framework forces — what spacetime geometry is made of if it is the deposited potential half of every decoherence event in cosmic history — we leave open.

It is the question that follows from taking the result seriously, and it is not answered by the data that exists today.

What is answered by the data is whether the sector split is real. Euclid will measure μ_0 to $\sigma(\mu_0) \approx 0.04$ in October 2026. The prediction is on record: $\mu_0 = -0.136$, $\Sigma_0 = 0$.

Data Availability

All MCMC chains, analysis scripts, modified CAMB Fortran source, and computational notebooks supporting this paper are publicly available at <https://github.com/hmahaffeyges/IAM-Validation> (Zenodo DOI: 10.5281/zenodo.18702042).

Acknowledgements

The author thanks the Planck, DESI, KiDS, DES, and HSC collaborations for making their data publicly available. This work made use of CAMB v1.5.8 [Lewis & Bridle, 2002], MGCAMB v1.5.2 [Wang et al., 2023], Cobaya [Torrado & Lewis, 2021], NumPy [Harris et al., 2020], SciPy [Virtanen et al., 2020], and Matplotlib [Hunter, 2007].

References

- Borghi, N., et al. 2022, ApJL, 928, L4
- Cai, R.-G. & Kim, S. P. 2005, JHEP, 02, 050
- DESI Collaboration, Adame, A. G., et al. 2025a, JCAP, 01, 124
- DESI Collaboration, Abdul-Karim, M., et al. 2025b, Phys. Rev. D, 112, 083515
- DESI Collaboration, Adame, A. G., et al. 2025c, JCAP, 07, 028
- Euclid Collaboration, Blanchard, A., et al. 2020, A&A, 642, A191
- Farooq, O. & Ratra, B. 2013, ApJ, 764, L46
- Harris, C. R., et al. 2020, Nature, 585, 357
- Hunter, J. D. 2007, Comput. Sci. Eng., 9, 90
- Jacobson, T. 1995, Phys. Rev. Lett., 75, 1260
- Jesus, J. F., et al. 2020, MNRAS, 514, 1403
- Landauer, R. 1961, IBM J. Res. Dev., 5, 183
- Lewis, A. & Bridle, S. 2002, Phys. Rev. D, 66, 103511
- Liu, G., Wang, Y., & Zhao, W. 2024, arXiv:2407.04385
- Mahaffey, H. W. 2026a, preprint. Code: <https://github.com/hmahaffeyges/IAM-Validation>. doi:10.5281/zenodo.18702042
- Mahaffey, H. W. 2026b, The Virial Partition Across Wide Range of Physical Scales: Cross-Domain Validation of the Informational Actualization Model. doi:10.5281/zenodo.18702042
- Mahaffey, H. W. 2026c, Dark Energy or Sector Tension? IAM Confrontation with DESI Full-Shape Growth Rates and Joint Weak Lensing. doi:10.5281/zenodo.18702042
- Mamon, A. A., Bamba, K., & Das, S. 2017, Eur. Phys. J. C, 77, 729
- Moresco, M. 2015, MNRAS, 450, L16
- Moresco, M., et al. 2012, JCAP, 08, 006
- Moresco, M., et al. 2016, JCAP, 05, 014
- Moresco, M., et al. 2022, Living Rev. Rel., 25, 6
- Ó Colgáin, E., et al. 2025, preprint (arXiv:2504.04417)
- Planck Collaboration, Aghanim, N., et al. 2020, A&A, 641, A6
- Riess, A. G., et al. 2004, ApJ, 607, 665
- Ratsimbazafy, A. L., et al. 2017, MNRAS, 467, 3239

Simon, J., Verde, L., & Jimenez, R. 2005, Phys. Rev. D, 71, 123001

Stern, D., et al. 2010, JCAP, 02, 008

Tomasetti, E., et al. 2023, A&A, 679, A96

Torrado, J. & Lewis, A. 2021, JCAP, 05, 057

Virtanen, P., et al. 2020, Nature Methods, 17, 261

Wang, Z., et al. 2023, JCAP, 2023, 022

Zhang, C., et al. 2014, RAA, 14, 1221